

Exp 1- Interfacing with Arduino

Project Name

Digital Temperature Monitoring System

Objective / Problem Statement

This project monitors temperature using a DHT11 sensor and displays live temperature data on an SSD1306 OLED display. The system can be turned ON/OFF using a push button. Based on the temperature value, LEDs, fan, and heater are controlled automatically.

Components Used

Component	Quantity
Arduino UNO	1
DHT11 Temperature Sensor	1
SSD1306 OLED Display	1
Red LED	1
Green LED	1
DC Fan / Motor	1
Heater / Bulb	1
Push Button	1
Jumper Wires	Required

---|---| Arduino UNO | 1 | | DHT11 / DHT22 Sensor | 1 | | SSD1306 OLED Display | 1 | | Buzzer | 1 | | Push Button | 1 | | Motor / Relay Module | 1 | | Jumper Wires | Required |

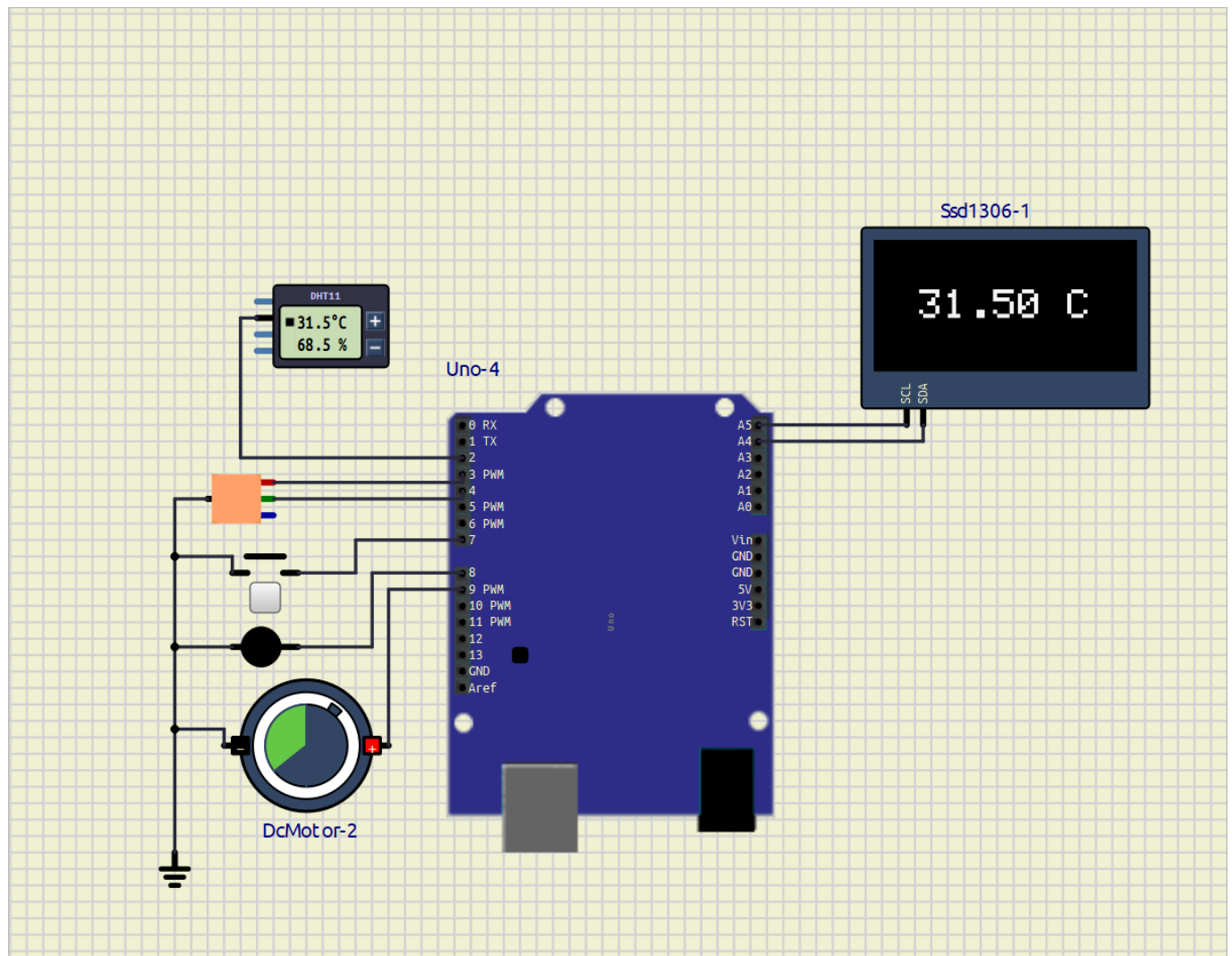
Pin Connections

Device	Arduino Pin
DHT11 Data Pin	D2
Red LED	D3
Green LED	D4
Push Button	D7
Heater	D8

Device	Arduino Pin
Fan	D9
OLED SDA	A4
OLED SCL	A5

---|---| DHT Sensor Data | D2 || OLED SDA | A4 || OLED SCL | A5 || Buzzer | D8 || Push Button | D7 || Motor / Relay | D9 |

Circuit Diagram



Required Libraries

```
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#include <DHT.h>
```

Main Arduino Code

```
// Include required libraries
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#include <DHT.h>

// OLED display width
#define SCREEN_WIDTH 128

// OLED display height
#define SCREEN_HEIGHT 64

// DHT11 data pin
#define DHTPIN 2

// DHT sensor type
#define DHTTYPE DHT11

// Create DHT object
DHT dht(DHTPIN, DHTTYPE);

// Create OLED display object
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);

// Variable to store ON/OFF state
bool systemON = false;

void setup()
{
  // Start DHT sensor
  dht.begin();

  // Initialize OLED display
  display.begin(SSD1306_SWITCHCAPVCC, 0x3C);

  // Clear OLED
  display.clearDisplay();

  // Red LED pin
  pinMode(3, OUTPUT);

  // Green LED pin
  pinMode(4, OUTPUT);

  // Fan pin
  pinMode(9, OUTPUT);

  // Heater pin
  pinMode(8, OUTPUT);
}
```

```
// Push button pin with internal pull-up resistor
pinMode(7, INPUT_PULLUP);
}

void loop()
{
    // Check if button is pressed
    if(digitalRead(7) == LOW)
    {
        // Toggle system state
        systemON = !systemON;

        // Small delay to prevent multiple toggles
        delay(300);
    }

    // If system is ON
    if(systemON)
    {
        // Read temperature
        float temp = dht.readTemperature();

        // Clear display
        display.clearDisplay();

        // Set text size
        display.setTextSize(2);

        // Set text color
        display.setTextColor(WHITE);

        // Create temperature text
        String text = String(temp) + " C";

        // Calculate center X position
        int x = (128 - (text.length() * 12)) / 2;

        // Calculate center Y position
        int y = (64 - 16) / 2;

        // Set cursor
        display.setCursor(x, y);

        // Print temperature
        display.print(text);

        // Update OLED
        display.display();

        // Temperature greater than or equal to 28
        if(temp >= 28)
        {
            // Turn ON red LED
            digitalWrite(3, HIGH);
        }
    }
}
```

```
// Turn OFF green LED
digitalWrite(4, LOW);

// Turn ON fan
digitalWrite(9, HIGH);

// Turn OFF heater
digitalWrite(8, LOW);
}
else
{
    // Turn OFF red LED
    digitalWrite(3, LOW);

    // Turn ON green LED
    digitalWrite(4, HIGH);

    // Turn OFF fan
    digitalWrite(9, LOW);

    // Turn ON heater
    digitalWrite(8, HIGH);
}
}

// If system is OFF
else
{
    // Turn OFF red LED
    digitalWrite(3, LOW);

    // Turn OFF green LED
    digitalWrite(4, LOW);

    // Turn OFF fan
    digitalWrite(9, LOW);

    // Turn OFF heater
    digitalWrite(8, LOW);

    // Clear OLED
    display.clearDisplay();

    // Update OLED
    display.display();
}

// Loop delay
delay(1000);
}
```

Working Principle

1. Push button toggles the system ON/OFF.
 2. DHT11 sensor reads room temperature.
 3. OLED display shows live temperature at the center of the screen.
 4. If temperature is greater than or equal to 28°C:
 - Red LED turns ON
 - Fan turns ON
 - Heater turns OFF
 5. If temperature is below 28°C:
 - Green LED turns ON
 - Heater turns ON
 - Fan turns OFF
 6. When the system is OFF, all devices remain OFF.
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Output

- OLED displays current temperature.
 - Red LED indicates high temperature.
 - Green LED indicates low temperature.
 - Fan runs automatically at high temperature.
 - Heater runs automatically at low temperature.
 - Push button controls complete system power state.
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Applications

- Smart room temperature control
 - Automatic cooling and heating system
 - Home automation projects
 - Industrial temperature monitoring
 - Embedded system learning projects
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